21·12·2022 (M)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Examinations
December 2022

Max. Marks: 100 | **Duration: 3 hrs**
Class: L Y | Semester: VII
Name of the Course: Computer Simulation and Modeling
Branch: Computer
Course Code: 2UCE718

Instructions:
(1) **All Questions are Compulsory**
(2) **Draw neat diagrams**
(3) **Assume suitable data if necessary**

Que. No.	Question	Max. Marks
Q1	Solve any **Four**	**20**
i)	When simulation is the appropriate tool ? Discuss the situation.	5
ii)	Write components of a typical simulation model.	5
iii)	Discuss model taxonomy.	5
iv)	Draw flow chart of calibration, verification and validation. Explain each block in the flow chart.	5
v)	How histograms can be used in the phase of input modeling ?	5
vi)	Discuss terminating and non-terminating simulation with respect to output analysis.	5

Que. No.	Question	Max. Marks
Q2 A	Solve the following	**10**
i)	Explain flow chart of execution of arrival event with respect to event scheduling algorithm.	5
ii)	With respect to event scheduling algorithm, define the terms: System state, Event, event list, Future event list, clock.	5
	OR	
Q2 A	Explain time advance algorithm. Tabulate clock, system state, future event list and cumulative statistics for single server queuing system with arrival and service details Inter-arrival times: **3 2 6 2 4 5** Service times: **2 5 5 8 4 5** Stop simulation when the clock reaches 20.	10
Q 2 B	Solve any **One**	**10**
i)	The maximum inventory level, M, is11 units and the review period, N, is 5 days. The problem is to estimate, by simulation, the average ending units in inventory and the number of days when a shortage condition occurs. Distribution of the number of units demanded per day is given.	10

Demand	Probability
0	0.10
1	0.25
2	0.35
3	0.21
4	0.09

Lead time distribution is given below.

Lead Time (Days)	Probability
1	0.6
2	0.3
3	0.1

Random digits for demand are; 24, 35, 65, 81, 54, 03, 87, 27, 73, 70, 47, 45, 48, 17, 09.

Random digits for lead time are; 5, 0, 3.

Assume that the simulation has been started with the inventory level at 3 units and an order of 8 units scheduled to arrive in 2 days' time.

ii) A computer technical support center is staffed by two people – Able and Baker who take calls and try to answer questions and solve computer problems. The time between calls ranges from 1 to 4 minutes, with distribution as given below. 10

Inter-arrival time (mins)	Probability
1	0.25
2	0.40
3	0.20
4	0.15

Able is more experienced and can provide service faster than Baker. The distributions of their service times are shown in following tables.

Distribution of Able's Service Time

Service time (mins)	Probability
2	0.30
3	0.28
4	0.25
5	0.17

Distribution of Baker's Service Time

Service time (mins)	Probability
3	0.35
4	0.25
5	0.20
6	0.20

When both are idle, Able takes the call. If both are busy, the customer has to wait. Simulate the system for 15 customers and compute average inter-arrival time, average service time of Able, average service time of Baker, average waiting time of customers, probability of idle time of Able, probability of idle time of Baker.

The random numbers for inter-arrival time are: 26, 98, 90, 26, 42, 74, 80, 68, 22, 48, 34, 45, 24, 34.

The random numbers for service time are: 95, 21, 51, 92, 89, 38, 13, 61, 50, 49, 39, 53, 88, 01, 81.

Que. No.	Question	Max. Marks
Q3	Solve any Two	20
i)	a) Explain geometric and negative binomial distribution. Write pmf of each distribution. b) Computer repair person is beeped each time if there is call for service. The number of beeps per hour follow Poisson distribution with $\alpha = 2$ per hour. Find The probability of exact three beeps in the next hour and the probability of two or more beeps in a 1-hour period.	10
ii)	A lamp has exponential distribution with $\lambda = 1/3$ per hour. On average, 1 failure per 3 hours.	10

	a) What is the probability that the lamp lasts longer than its mean life? b) The probability that the lamp lasts between 2 to 3 hours? c) The probability that it lasts for another hour given it is operating for 2.5 hours?	
iii)	Two workers competing for a job, Able claims to be faster than Baker on average, but Baker claims to be more consistent. The arrivals are Poisson arrivals at rate 2 per hour (1/30 per minute). The Able's mean service time is 24 minutes with variance of 400 minutes^2. The Baker's mean service time is 25 minutes with variance of 4 minutes^2 a) Find out which worker will have a larger waiting queue. b) Find out the probability of having 0 customers in Able and Baker's queue.	10

Que. No.	Question	Max. Marks
Q4	Solve any **Two**	**20**
i)	What is the role of maximum density period in generation of random numbers? With given seed 45, constant multiplier 21, increment 49 and modulus 40, generate a sequence of five random numbers.	10
ii)	Using suitable frequency test find out whether the random numbers generated are uniformly distributed on the interval [0,1] can be rejected. Assume α=0.05 and $D\alpha$=0.565.the random numbers are 0.54, 0.73, 0.98, 0.11, 0.68	10
iii)	Explain inverse transform technique of producing random variates for exponential distribution. Can inverse transform technique be used for generating random variates of all distribution ? Jutsify the answer.	10

Que. No.	Question	Max. Marks
Q5	Write notes on any **four**	**20**
i)	M/M/1 Queuing model	5
ii)	Properties of Random numbers	5
iii)	Estimation of performance in output analysis	5
iv)	Selecting the distributions in input modeling phase	5
v)	Simulation Tools	5
vi)	Applications of simulation in manufacturing	5